Animation inside a Jupyter notebook

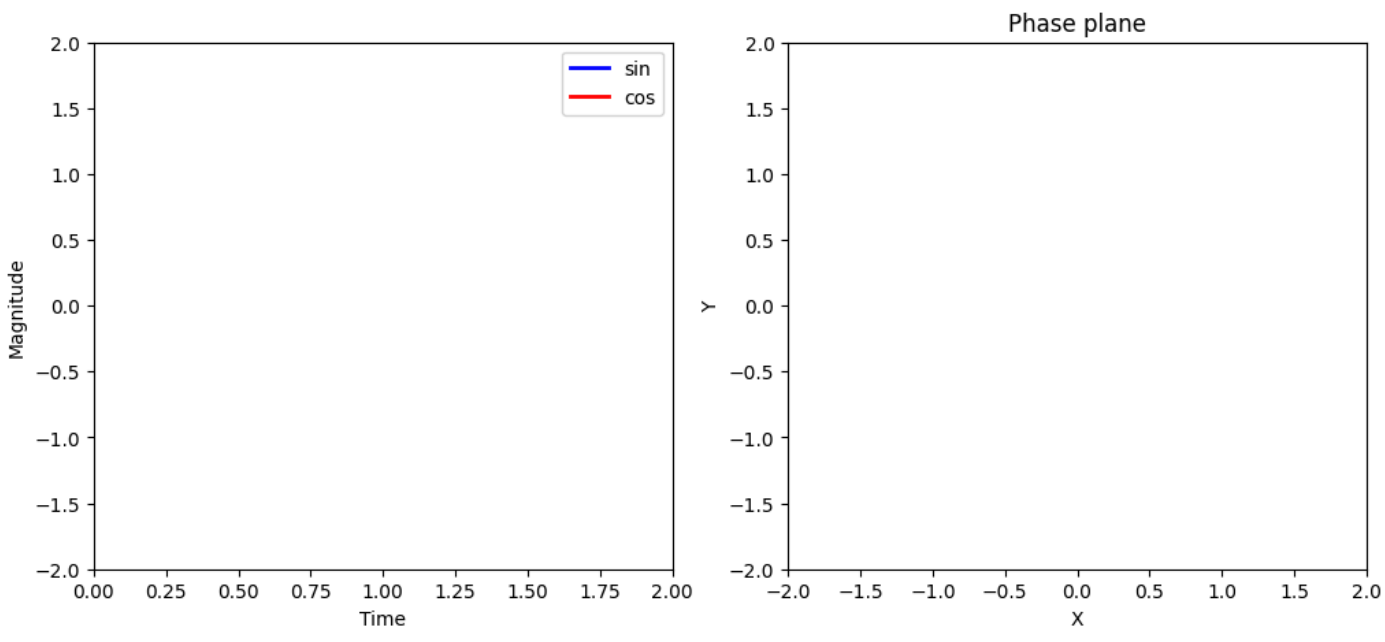
Contents

Next to (plain or enhanced) MarkDown, Jupyter books can also contain chapters which are essentially Jupyter Notebooks. The current file is an example, in which I've created an animation in a Jupyter Notebook, which plots the sine/cosine and their phase plane over time.

Source: colab Google

6.1. Step 1: Background frame

```
%matplotlib inline

import numpy as np
import matplotlib.pyplot as plt

# Create a figure and axes.
fig = plt.figure(figsize=(12,5))
ax1 = plt.subplot(1,2,1)
ax2 = plt.subplot(1,2,2)

# Set up the subplots
ax1.set_xlim((0,2))
ax1.set_ylim((-2,2))
ax1.set_xlabel('Time')
ax1.set_ylabel('Magnitude')

ax2.set_xlim((-2,2))
ax2.set_ylim((-2,2))
ax2.set_xlabel('X')
ax2.set_ylabel('Y')
ax2.set_title('Phase plane')

# Create objects that will change in the animiation.
# These objects are initially empty, and will be given new values for each frame in the animation.
txt_title = ax1.set_title('')
line1, = ax1.plot([], [], 'b', lw=2)  # ax.plot returns a list of 2D line objects.
line2, = ax1.plot([], [], 'r', lw=2)
pt1, = ax2.plot([], [], 'g.', ms=20)
line3, = ax2.plot([], [], 'y', lw=2)

ax1.legend(['sin', 'cos']);
```



6.2. Step 2: Define a function to draw each frame

```
# Animation function. This function is called sequentially.
def drawframe(n):
    x = np.linspace(0, 2, 1000)
    y1 = np.sin(2 * np.pi * (x - 0.01 * n))
    y2 = np.cos(2 * np.pi * (x - 0.01 * n))
    line1.set_data(x, y1)
    line2.set_data(x, y2)
    line3.set_data(y1[0:50],y2[0:50])
    pt1.set_data([y1[0]], [y2[0]])   # Note that matplotlib will throw an error if we supply only numbers (i.e., p
    txt_title.set_text('Frame = {0:4d}'.format(n))
    return (line1,line2)
```

```
# Initialization function.
def init():
    line1.set_data([],[])
    line2.set_data([],[])
    return(line1, line2)
```

6.3. Step 3: Create the animation object

```
from matplotlib import animation

#anim = animation.FuncAnimation(fig, drawframe, init_func=init, frames=100, interval=20, blit=True)
anim = animation.FuncAnimation(fig, drawframe, frames=100, interval=20, blit=True)
# blit = True re-draws only the parts that have changed.
```

6.4. Step 4 (JavaScript version)

Use HTML(anim.to_jshtml()) See Stackoverflow inline animations in Jupyter. Note that this takes some time to parse.

```
from IPython.display import HTML
HTML(anim.to_jshtml())
```